electron has a definite position in the usual intuitive sense only at the moment we “look” at it—at the moment when we measure its position—identifying its location with certainty. But before (and after) we do that, all it has are potential positions described by a probability wave that, like any wave, is subject to interference effects. It’s not that the electron has a position and that we don’t know the position before we do our measurement. Rather, contrary to what you’d expect, the electron simply *does not have* a definite position before the measurement is taken.

This is a radically strange reality. In this view, when we measure the electron’s position we are not measuring an objective, preexisting feature of reality. Rather, the act of measurement is deeply enmeshed in creating the very reality it is measuring. Scaling this up from electrons to everyday life, Einstein quipped, “Do you really believe that the moon is not there unless we are looking at it?” The adherents of quantum mechanics responded with a version of the old saw about a tree falling in a forest: if no one is looking at the moon—if no one is “measuring its location by seeing it”—then there is no way for us to know whether it’s there, so there is no point in asking the question. Einstein found this deeply unsatisfying. It was wildly at odds with his conception of reality; he firmly believed that the moon is there, whether or not anyone is looking. But the quantum stalwarts were unconvinced.

Einstein’s second challenge, raised at the Solvay conference in 1930, followed closely on the first. He described a hypothetical device, which (through a clever combination of a scale, a clock, and a cameralike shutter) seemed to establish that a particle like an electron *must* have definite features—before it is measured or examined—that quantum mechanics said it couldn’t. The details are not essential but the resolution is particularly ironic. When Bohr learned of Einstein’s challenge, he was knocked back on his heels—at first, he couldn’t see a flaw in Einstein’s argument. Yet, within days, he bounced back and fully refuted Einstein’s claim. And the surprising thing is that the key to Bohr’s response was general relativity! Bohr realized that Einstein had failed to take account of his own discovery that gravity warps time—that a clock ticks at a rate dependent on the gravitational field it experiences. When this complication was included, Einstein was forced to admit that his conclusions fell right in line with orthodox quantum theory.